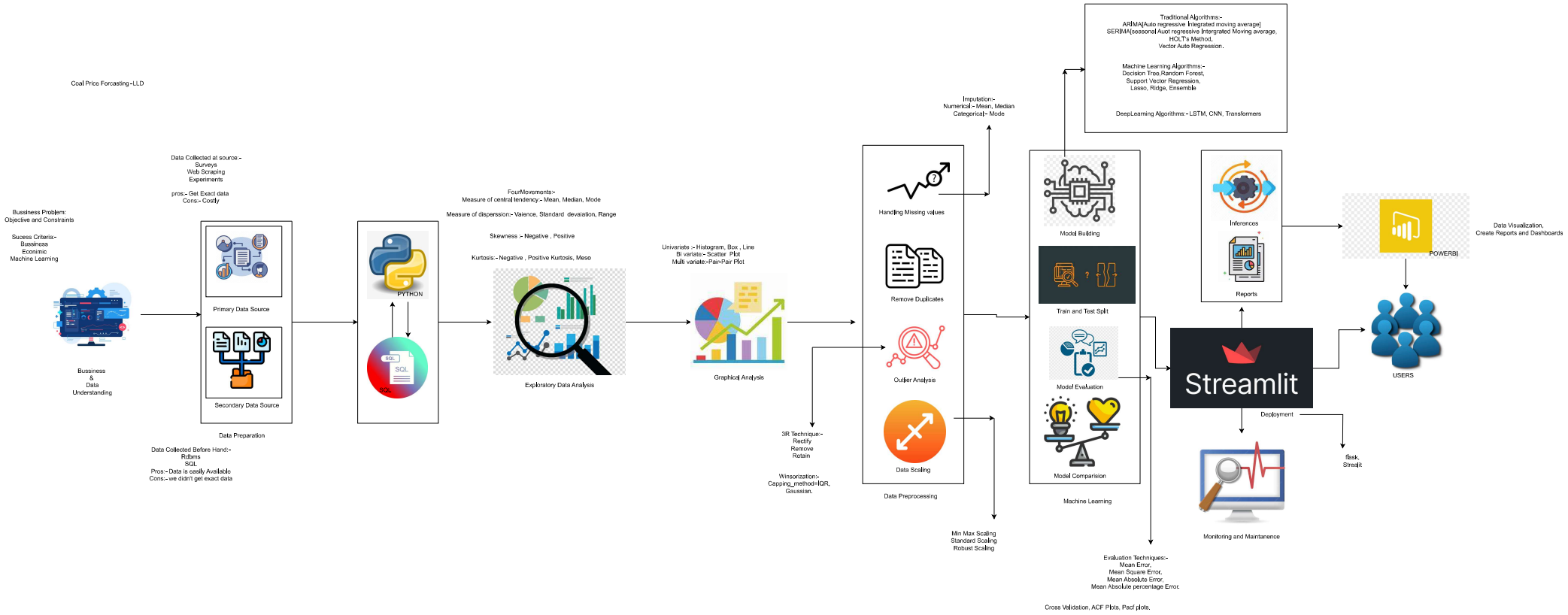
Coal Price Forcasting - LLD
Bussiness Problem:
Objective and Constraints
Sucess Criteria:-
Bussiness
Econimic
Machine Learning
Bussiness
&
Data
Understanding
Data Collected at source:-
Surveys
Web Scraping
Experiments
pros:- Get Exact data
Cons - Costly
Primary Data Source
Secondary Data Source
Data Preparation
Data Collected Before Hand:-
Robins
SQL
Pros:- Data is easily Available
Cons:- we didn't get exact data
PYTHON
SQL
FourMovements:-
Measure of central tendency - Mean, Median, Mode
Measure of dispersion:- Vaience, Standard devaiation, Range
Skewness :- Negative , Positive
Kurtosis:- Negative , Positive Kurtosis, Meso
Exploratory Data Analysis
Univariate :- Histogram, Box , Line
Bi variate:- Scatter Plot
Multi variate:-Pair-Pair Plot
Graphical Analysis
Handling Missing values
Remove Duplicates
Outlier Analysis
Data Scaling
Data Preprocessing
3R Technique:-
Rectify
Remove
Retain
Winsorization:-
Capping_method=IQR,
Gaussian.
Imputation:-
Numerical - Mean, Median
Categorical:- Mode
Min Max Scaling
Standard Scaling
Robust Scaling
Traditional Algorithms:-
ARIMA[Auto regressive Integrated moving average]
SERIMA[seasonal Auot regressive Intergrated Moving average,
HOLT's Method,
Vector Auto Regression.
Machine Learning Algorithms:-
Decision Tree,Random Forest,
Support Vector Regression,
Lasso, Ridge, Ensemble
DeepLearning Algorithms:- LSTM, CNN, Transformers
Model Building
Train and Test Split
Model Evaluation
Model Comparision
Machine Learning
Evaluation Techniques:-
Mean Error,
Mean Square Error,
Mean Absolute Error,
Mean Absolute percentage Error.
Cross Validation, ACF Plots, Pacf plots.
Inferences
Reports
Streamlit
Deployment
flask,
Streamlit
Monitoring and Maintanence
POWERBI
Data Visualization,
Create Reports and Dashboards
USERS